

Utility-Driven Machine Learning-Based Adaptive Link Selection for Weather-Aware Hybrid VLC–RF Vehicular Platooning

Abstract—Vehicular platooning in next-generation intelligent transportation systems requires ultra-reliable, low-latency, and high-throughput vehicle-to-vehicle (V2V) communication under dynamic weather and mobility conditions. This paper proposes a utility-driven machine learning-based adaptive link aggregation framework for heterogeneous Visible Light Communication (VLC) and Radio Frequency (RF) vehicular platooning systems. Unlike conventional binary RF/VLC switching, the proposed framework enables adaptive selection among standalone RF, standalone VLC, and link-aggregated transmission modes. A Monte Carlo simulation framework is developed to generate 60,000 samples incorporating realistic mobility, traffic density, rain, and fog impairments. A multi-objective utility function jointly optimizing throughput, delay, and outage probability is formulated, and optimal utility weights are obtained through 10-fold cross-validated grid search optimization, yielding the optimal weight vector $(w_C, w_D, w_O) = (0.7, 0.2, 0.1)$. A Random Forest classifier is trained for real-time link selection and achieves 89.71% accuracy. Simulation results demonstrate that the proposed link-aggregated mode achieves up to 38.6% higher utility under dense fog and nearly one order of magnitude delay reduction compared with standalone VLC at longer distances. Feature importance analysis identifies fog attenuation, inter-vehicle distance, and weather class as dominant decision factors. The proposed framework enables scalable and weather-aware intelligent link selection for autonomous vehicular platooning systems.

Index Terms—Visible Light Communication, Radio Frequency, Link-Aggregated V2X, Link Selection, Random Forest, Optimization, Multi-Objective Utility, Vehicular Networks

I. INTRODUCTION

Vehicular platooning is a key component of next-generation intelligent transportation systems, enabling cooperative driving with reduced inter-vehicle spacing, improved road capacity, and enhanced safety. Such coordination requires reliable, low-latency, and high-throughput vehicle-to-everything (V2X) communication for ubiquitous information exchange and platoon stability [1]. Radio Frequency (RF) communication at 5.9 GHz through dedicated short range communication (DSRC) and cellular-V2X (C-V2X) provides robust long-range connectivity under non-line-of-sight (NLoS) conditions; however, dense platooning scenarios introduce spectrum congestion, co-channel interference, and channel contention, degrading RF performance [2]. These challenges motivate the integration of complementary communication technologies for improved QoS support.

Visible Light Communication (VLC), using vehicle LED headlights and taillights, provides high-bandwidth,

interference-free communication under line-of-sight conditions, achieving data rates of tens of Mbps with inherent physical-layer security due to its directional nature [3]. Combined with Radio Frequency (RF) systems, it enables link-aggregated VLC–RF architectures where VLC supports short-range, high-capacity platoon communication while RF ensures reliable long-range connectivity, with aggregation further enhancing diversity and overall performance [4]. However, real-time adaptive selection of RF, VLC, and aggregated modes remains challenging due to dynamic channel conditions and mobility. Existing threshold and optimization-based methods rely on simplifying assumptions [5], while current learning-based approaches are still limited to partial decision spaces and struggle to fully exploit complementary links, reducing effectiveness in practical platooning scenarios [6]. These limitations motivate more adaptive and intelligent link selection strategies for robust vehicular communication.

Most of the existing studies on VLC–RF vehicular systems mainly focus on binary RF/VLC switching or optimize a single metric such as SNR or capacity, which limits their effectiveness in realistic platooning scenarios with multiple conflicting QoS requirements. *The novelty of this work lies in (i) the development of a unified utility-driven framework that jointly optimizes throughput, delay, and outage probability for balanced QoS-aware link selection, (ii) the integration of RF, VLC, and link aggregation into a single adaptive decision space to dynamically exploit their complementary communication characteristics, and (iii) the introduction of a utility-driven machine learning-based link selection strategy that learns optimal transmission policies from data instead of relying on heuristic switching rules, thereby improving adaptability and robustness in hybrid VLC–RF vehicular platooning networks.*

II. SYSTEM MODEL AND PRELIMINARIES

We consider a link-aggregated VLC–RF vehicular platooning system operating on a single-lane highway segment, as illustrated in Fig. 1. Vehicles are spatially distributed according to a Poisson Point Process (PPP) with intensity λ_v vehicles/m, yielding exponentially distributed inter-vehicle spacings. Each vehicle is equipped with:

- An RF transceiver operating at 5.9 GHz (DSRC/C-V2X) with transmit power $P_t^{RF} = 20$ dBm and bandwidth $B_{RF} = 10$ MHz.

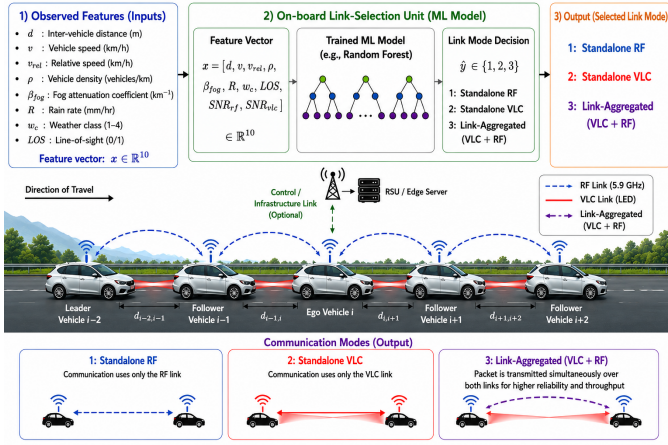


Fig. 1: System model of the proposed machine learning-based utility-driven link aggregation framework for VLC-RF vehicular platooning under adverse weather conditions.

- A VLC transceiver using LED headlights/tailights with optical transmit power $P_t^{VLC} = 10$ W, responsivity $R_d = 0.4$ A/W, detector area $A = 10^{-4}$ m², and bandwidth $B_{VLC} = 20$ MHz.
- An on-board link selector (the trained ML model) that observes the feature vector and outputs the optimal link mode $\hat{y} \in \{1, 2, 3\}$ (RF, VLC, Link-Aggregated).

A. Communication channel models

The RF received power is modeled using the log-distance path loss model with log-normal shadowing [7]:

$$P_r^{\text{RF}} (\text{dBm}) = P_t^{\text{RF}} - PL_0 - 10n \log_{10}(d) - \xi \quad (1)$$

where $PL_0 = 20 \log_{10}(4\pi/\lambda_c)$ denotes the free-space reference path loss at carrier wavelength λ_c (5.9 GHz), $n = 2.2$ is the vehicular highway path loss exponent, and $\xi \sim \mathcal{N}(0, \sigma_s^2)$ with $\sigma_s = 4$ dB models log-normal shadowing. The effective RF SNR is computed as [7]:

$$\gamma_{\text{RF}} = \frac{P_r^{\text{RF}, \text{lin}}}{N_{\text{RF}}^{\text{lin}}} \prod_{k=1}^4 \Pi_k^{\text{RF}} \quad (2)$$

where $N_{\text{RF}} = -96$ dBm denotes the receiver noise floor and Π_k^{RF} represent the RF channel impairment penalties corresponding to rain attenuation, gaseous absorption, co-channel interference, and Doppler fading.

The achievable RF capacity and outage probability are given by $C_{\text{RF}} = B_{\text{RF}} \log_2(1 + \gamma_{\text{RF}})$ and $P_{\text{out}}^{\text{RF}} = 1 - e^{-\gamma_{\text{th}}/\gamma_{\text{RF}}}$, respectively, where the outage threshold is defined as $\gamma_{\text{th}} = 2^{C_{\text{min}}/B_{\text{RF}}} - 1$ with minimum required rate $C_{\text{min}} = 2$ Mbps. The rain-induced RF attenuation penalty is modeled as $\Pi_{\text{rain}}^{\text{RF}} = \exp(-0.06R)$, where R denotes the rain rate in mm/hr.

For aligned vehicular platooning scenarios, the LOS VLC channel gain based on the Lambertian radiation model is expressed as [13]:

$$H_{\text{LOS}} = \frac{(m+1)A}{2\pi d^2} \cos^m(\phi) T_s(\psi) g(\psi) \cos(\psi) \quad (3)$$

where m denotes the Lambertian emission order, A is the photodetector area, d is the inter-vehicle distance, ϕ is the irradiance angle, ψ is the incidence angle, $T_s(\psi)$ is the optical filter gain, and $g(\psi)$ is the optical concentrator gain.

Weather-dependent attenuation due to fog and rainfall is modeled using the Beer-Lambert law as [8]:

$$H_{\text{VLC}} = H_{\text{LOS}} \exp(-18.0 \beta_{\text{fog}} d_{\text{km}}) \exp(-0.01 R d_{\text{km}}) \quad (4)$$

where β_{fog} is the fog extinction coefficient (km⁻¹), R denotes the rain rate in mm/hr, and $d_{\text{km}} = d/1000$ is the inter-vehicle distance in kilometers.

LOS blockage caused by vehicle density is modeled probabilistically as

$$P_{\text{LOS}}(d, \lambda_v) = \exp(-k \lambda_v d / 1000) \quad (5)$$

where λ_v denotes the vehicle density and $k = 0.05$. The blockage-dependent attenuation factor is then assigned according to the LOS probability as

$$\eta_{\text{LOS}} = \begin{cases} 0.6, & \text{with probability } P_{\text{LOS}}(d, \lambda_v) \\ 0.1, & \text{with probability } 1 - P_{\text{LOS}}(d, \lambda_v). \end{cases} \quad (6)$$

The effective VLC SNR incorporating mobility-induced alignment jitter is expressed as [9]

$$\gamma_{\text{VLC}} = \frac{(R_d P_t^{VLC} H_{\text{VLC}} \eta_{\text{LOS}})^2}{\sigma_{\text{VLC}}^2} \exp(-v_{\text{rel}}/80) \quad (7)$$

where R_d is the photodetector responsivity, $\sigma_{\text{VLC}}^2 = 10^{-9}$ denotes the receiver noise variance, and v_{rel} is the relative vehicle speed.

The achievable VLC capacity and outage probability are given by $C_{\text{VLC}} = B_{\text{VLC}} \log_2(1 + \gamma_{\text{VLC}})$ and $P_{\text{out}}^{\text{VLC}} = \Pr(\gamma_{\text{VLC}} < \gamma_{\text{th}})$, respectively, where the outage threshold is defined as $\gamma_{\text{th}} = 2^{C_{\text{min}}/B_{\text{VLC}}} - 1$ with minimum required rate $C_{\text{min}} = 2$ Mbps.

B. Link-aggregated VLC-RF

The Link-Aggregated mode splits the packet load across both RF and VLC channels concurrently, where the load allocation factor α is determined by the relative channel capacities as $\alpha = \frac{C_{\text{RF}}}{C_{\text{RF}} + C_{\text{VLC}} + \epsilon}$ [9], with $\epsilon = 10^{-12}$ included to prevent division-by-zero conditions. For the link-aggregated transmission mode, the load-split delay is computed as:

$$D_{\text{LA}} = \max\left(\frac{\alpha L}{C_{\text{RF}}}, \frac{(1-\alpha)L}{C_{\text{VLC}}}\right) \cdot (1 + \Delta_{\text{imb}}) \quad (8)$$

where $L = 1500 \times 8 = 12,000$ bits is the packet size and $\Delta_{\text{imb}} = |C_{\text{RF}} - C_{\text{VLC}}|/(C_{\text{RF}} + C_{\text{VLC}})$ is a packet-reordering penalty for channel imbalance [10]. The Link-aggregated effective capacity is:

$$C_{\text{LA}} = 0.6 \cdot (C_{\text{RF}} + C_{\text{VLC}}) \quad (9)$$

The 0.6 factor accounts for MAC synchronization overhead as reported in [9].

III. DATASET GENERATION AND UTILITY OPTIMIZATION

A. Monte Carlo-Based Dataset Generation

Monte Carlo simulation is employed to generate a statistically diverse dataset capturing mobility, traffic density, weather impairments, interference, and line-of-sight (LOS) blockage in heterogeneous VLC–RF vehicular environments. Owing to the absence of publicly available datasets that simultaneously capture hybrid VLC/RF communication characteristics, vehicular mobility dynamics, environmental variations, and communication-performance metrics for platooning scenarios, a Monte Carlo simulation-based dataset¹ is adopted to generate representative communication samples. Simulation-driven datasets are widely employed in vehicular networking research when large-scale labeled measurements are unavailable [14], [15]. A total of $N = 60,000$ samples are generated in MATLAB using a fixed random seed (`rng(1)`) to ensure reproducibility. The mobility parameters are modeled as: inter-vehicle distance $d = \min(10 + \exp(\lambda_v), 200)$ m, where $\exp(\lambda_v)$ denotes an exponential distribution with rate $\lambda_v = 0.015$ vehicles/m; vehicle speed $v \sim \mathcal{U}(0, 140)$ km/h, where $\mathcal{U}(a, b)$ denotes a uniform distribution over $[a, b]$; relative speed $v_{rel} = |30\mathcal{N}(0, 1)|$ km/h, where $\mathcal{N}(0, 1)$ denotes the standard normal distribution; and vehicle density $\rho \sim \mathcal{U}(20, 100)$ vehicles/km. The dataset is equally divided into four weather classes ($N_w = 15,000$ each): clear sky ($\beta_{fog} = 0, R = 0$), rain ($R \sim \mathcal{U}(5, 50)$ mm/hr), moderate fog ($\beta_{fog} \sim \mathcal{U}(1.5, 3)$ km⁻¹), and dense fog ($\beta_{fog} \sim \mathcal{U}(3, 6)$ km⁻¹).

B. Utility-based three-class labeling

For each link type $l \in \{RF, VLC, LA\}$, the channel capacity, delay, and outage probability are denoted by C_l , D_l , and $P_{out,l}$, respectively. The channel metrics are first normalized across all samples as $C_{l,n} = C_l/C_{max}$ and $D_{l,n} = D_l/D_{max}$, where C_{max} and D_{max} denote the global maximum capacity and delay values computed over all samples and link types. The utility score, U_l for each link is then computed as

$$U_l = w_C C_{l,n} - w_D D_{l,n} - w_O P_{out,l} \quad (10)$$

where w_C , w_D , and w_O represent the weighting coefficients associated with normalized capacity, delay, and outage probability, respectively. The final link label for sample i is assigned according to

$$y_i = \arg \max_{l \in \{RF, VLC, LA\}} U_l^{(i)} \quad (11)$$

where $U_{LA}^{(i)} = -\infty$ for samples in which simultaneous RF–VLC transmission is infeasible due to severe channel degradation or invalid aggregation conditions. The resulting label distribution over 60,000 samples is RF: 18,757 (31.3%), VLC: 16,219 (27.0%), and Link-Aggregated: 25,024 (41.7%). The relatively balanced class proportions avoid severe imbalance and ensure adequate representation of all communication

modes during training, while equal weather-wise sampling (15,000 samples per class) prevents environmental bias.

C. Grid-Search Optimization of Utility Weights

The proposed framework employs exhaustive grid-search with 10-fold cross-validation to optimize the utility weights for capacity, delay, and outage probability, aiming to achieve balanced selection among RF, VLC, and link-aggregated modes. The weights w_C and w_D are searched within $[0.1, 1.0]$ using a step size of 0.05, while the outage weight is computed as $w_O = 1 - (w_C + w_D)$ under the constraint $w_O \geq 0.1$. For each candidate weight set, the utilities of RF, VLC, and link-aggregated links are evaluated using (11), and the communication mode is selected according to

$$Label = \arg \max(U_{RF}, U_{VLC}, U_{LA}). \quad (12)$$

Algorithm 1 Grid Search for Optimal Utility Weight Selection

Require: Normalized capacities (C_{RF}, C_{VLC}, C_{LA}), delays (D_{RF}, D_{VLC}, D_{LA}), outage probabilities (P_{RF}, P_{VLC}, P_{LA})

Ensure: Optimal weights (w_C^*, w_D^*, w_O^*), best score, final labels

- 1: Define feasible weight set: $w_C + w_D + w_O = 1, w_i \geq 0$
 - 2: Initialize $best_score \leftarrow -\infty$
 - 3: **for** each candidate (w_C, w_D) in grid **do**
 - 4: $w_O \leftarrow 1 - (w_C + w_D)$
 - 5: **if** $w_O < 0$ **then**
 - 6: **continue**
 - 7: **end if**
 - 8: Compute utilities:
 - 9: $U_{RF} \leftarrow w_C C_{RF} - w_D D_{RF} - w_O P_{RF}$
 - 10: $U_{VLC} \leftarrow w_C C_{VLC} - w_D D_{VLC} - w_O P_{VLC}$
 - 11: $U_{LA} \leftarrow w_C C_{LA} - w_D D_{LA} - w_O P_{LA}$
 - 12: If infeasible, set $U_{LA} \leftarrow -\infty$
 - 13: $pred \leftarrow \arg \max(U_{RF}, U_{VLC}, U_{LA})$
 - 14: Compute score:
 - 15: $score \leftarrow -\text{std}(N_{RF}/N, N_{VLC}/N, N_{LA}/N)$
 - 16: **if** $score > best_score$ **then**
 - 17: $best_score \leftarrow score$
 - 18: Store (w_C, w_D, w_O) as best weights
 - 19: **end if**
 - 20: **end for**
 - 21: Compute final utilities using (w_C^*, w_D^*, w_O^*)
 - 22: Output final labels: RF / VLC / Link-Aggregated
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To avoid severe class imbalance, a balancing score based on the standard deviation of the class proportions is evaluated as

$$Score = -\sigma \left(\frac{N_{RF}}{N}, \frac{N_{VLC}}{N}, \frac{N_{LA}}{N} \right),$$

where $\sigma(\cdot)$ denotes the standard deviation operator, N_{RF} , N_{VLC} , and N_{LA} represent the number of samples assigned to RF, VLC, and Link-Aggregated classes, respectively, and N is the total number of samples. The final score is averaged over

¹The source code used to generate the dataset and reproduce the reported results is publicly available on GitHub at <https://11nq.com/3ktho8i>.

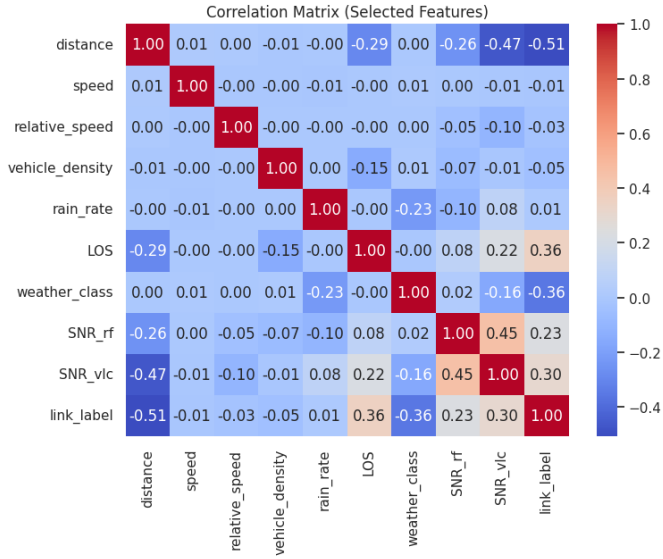


Fig. 2: Correlation matrix of selected features and the class label used for link selection.

$K = 10$ folds, and the weight set yielding the highest average score is selected as the optimal solution (w_C^*, w_D^*, w_O^*). The complete procedure is summarized in **Algorithm 1**.

IV. ML-BASED LINK SELECTION FRAMEWORK

A. Feature Engineering

The 10-dimensional feature vector for each training sample is:

$$\mathbf{x} = [d, v, v_{\text{rel}}, \rho, \beta_{\text{fog}}, R, w_{\text{class}}, \text{LOS}, \text{SNR}_{\text{rf}}, \text{SNR}_{\text{vlc}}]$$

comprising inter-vehicle distance (m), vehicle speed (km/h), relative speed (km/h), vehicle density (vehicles/km), fog extinction coefficient (km^{-1}), rain rate (mm/hr), $w_{\text{class}} \in \{1, 2, 3, 4\}$, and LOS indicator $\in \{0, 1\}$. This feature set captures the joint effects of mobility, traffic density, and atmospheric conditions that govern the comparative performance of RF and VLC channels. Fig. 2 shows correlation matrix of selected features and the class label used for link selection. It can be observed that inter-vehicle distance, weather conditions, LOS availability, and VLC SNR are the dominant factors influencing adaptive RF/VLC/link-aggregated mode selection, while mobility-related features exhibit comparatively weaker influence.

B. Random Forest Classifier

The Random Forest classifier is trained as an ensemble of $N_T = 100$ decision trees, where each tree is constructed using bootstrap sampling and feature subsampling $m_{\text{try}} = \lfloor \sqrt{10} \rfloor$ at each split node. The hyperparameters are configured as: maximum depth = 5, minimum samples per split = 10, and minimum samples per leaf = 8, with balanced class weights

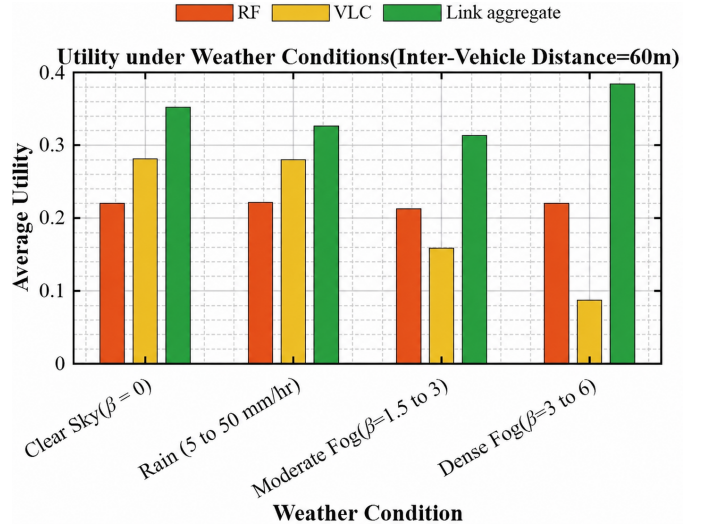


Fig. 3: Average utility comparison under four weather conditions at inter-vehicle distance ≈ 60 m.

to mitigate residual class imbalance. The final class prediction is obtained through majority voting:

$$\hat{y} = \arg \max_{c \in \{1, 2, 3\}} \frac{1}{N_T} \sum_{t=1}^{N_T} \mathbf{1}[f_t(\mathbf{x}) = c] \quad (13)$$

where $f_t(\mathbf{x})$ denotes the class predicted by the t -th decision tree for input feature vector $\mathbf{x}$. Gini-based feature importance analysis identifies *fog_beta* as the dominant feature due to VLC sensitivity to fog attenuation, followed by *distance* and *weather_class*, which strongly influence RF/VLC link preference. *vehicle_density* and *rain_rate* further affect utility through interference, LOS blockage, and weather-induced degradation, while *speed*, *relative_speed*, and *LOS* collectively capture mobility and link geometry effects.

V. SIMULATION RESULTS

We evaluate the proposed Link-Aggregated VLC–RF framework against conventional standalone RF and VLC communication under realistic vehicular conditions. While RF provides robust performance under adverse weather and non-line-of-sight conditions, VLC offers higher throughput under clear LOS scenarios. The proposed framework aims to exploit the complementary advantages of both technologies to achieve improved utility, lower delay, and enhanced reliability across diverse platooning environments. Performance is analyzed under four weather conditions: clear sky ($\beta_{\text{fog}} = 0$), rain ($R \in [5, 50]$ mm/hr), moderate fog ($1.5 \leq \beta_{\text{fog}} < 3.0 \text{ km}^{-1}$), and dense fog ($\beta_{\text{fog}} \geq 3.0 \text{ km}^{-1}$).

Fig. 3 presents the average utility comparison of RF, VLC, and the Link-aggregated modes across four weather classes at a fixed inter-vehicle distance of approximately 60 m. It can be observed that irrespective of weather conditions, the link-aggregated VLC–RF consistently achieves the highest utility in all four weather conditions. Under clear sky ($\beta = 0$) and light fog ($\beta = 0.3$ – 1.5) condition, standalone VLC performs

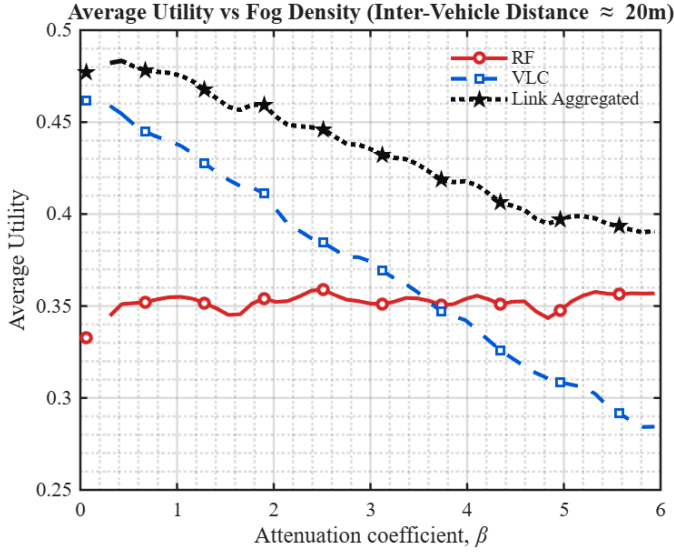


Fig. 4: Average utility vs. fog attenuation coefficient β at inter-vehicle distance ≈ 20 m.

better than standalone RF as expected. However, standalone RF outperforms standalone VLC under dense fog condition. Notably, the link-aggregated VLC-RF achieves approximately 38.6% higher average utility than standalone VLC under dense fog conditions.

Fig. 4 examines the variation of average utility as a function of the fog attenuation coefficient β at a fixed inter-vehicle separation of 20 m. As evident, the VLC link starts with high utility under clear conditions (utility ≈ 0.46) due to its high received power at short range, and decreases monotonically with increasing β , reaching ≈ 0.28 at $\beta = 6.0 \text{ km}^{-1}$. The RF link exhibits near-constant utility (≈ 0.35), largely unaffected by fog. The link-aggregated link maintains the highest utility across the entire fog range (≈ 0.39 – 0.48), exploiting VLC when it is beneficial and progressively offloading to RF as fog thickens. This behavior confirms that the Link-Aggregated mode acts as an adaptive umbrella: it inherits VLC's short-range capacity advantage under good visibility and falls back on RF's weather robustness as optical attenuation grows.

Fig. 5 shows the average utility of all three link modes as a function of inter-vehicle distance under clear weather ($\beta = 0, R = 0$). At short distances ($d < 30$ m), VLC achieves the highest single-link utility (≈ 0.52) due to its superior received power and high capacity. As distance increases, VLC utility decays rapidly due to the $1/d^2$ channel gain fall-off and alignment losses. RF decays more slowly due to its lower path-loss exponent effect at 5.9 GHz. The Link-Aggregated mode maintains the highest utility at all distances, with a diminishing but persistent advantage even at $d = 200$ m (≈ 0.12 vs. 0.07 for VLC and 0.02 for RF alone). This result confirms that link aggregation is beneficial across the full operational range, not only at short distances.

Fig. 6 presents the average packet delay on a logarithmic scale as a function of inter-vehicle distance at a fixed fog

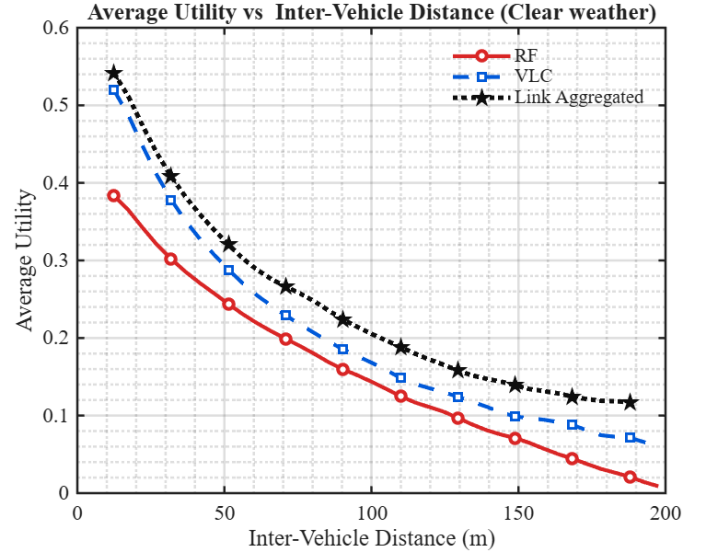


Fig. 5: Average utility vs. inter-vehicle distance under clear weather.

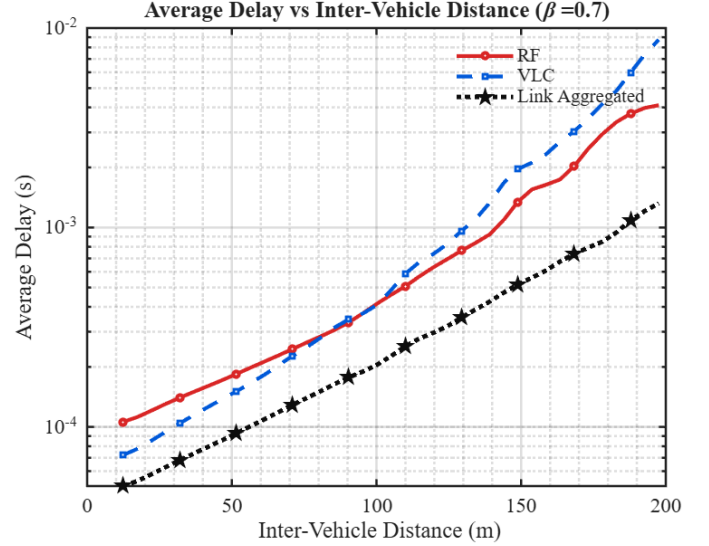


Fig. 6: Average delay (log scale) vs. inter-vehicle distance at $\beta \approx 0.7 \text{ km}^{-1}$.

level $\beta \approx 0.7 \pm 0.7 \text{ km}^{-1}$ (encompassing the light-to-moderate fog transition). At short distances, all three links exhibit low delay ($\approx 10^{-4}$ s). As distance increases, both RF and VLC delays increase monotonically due to degrading SNR and reduced capacity. The VLC delay escalates more steeply than RF, consistent with its faster channel gain decay. The Link-Aggregated link achieves the lowest delay across the entire distance range, approximately one order of magnitude lower than standalone VLC at $d = 200$ m ($\approx 10^{-3}$ s vs. $\approx 7 \times 10^{-3}$ s). This demonstrates the critical role of link aggregation in meeting the sub-millisecond latency requirements of safety-critical V2V applications.

Table I compares the Random Forest classifier trained on the optimized utility weights (0.7, 0.2, 0.1) against three fixed-

TABLE I: Performance Comparison: Fixed vs. Optimized utility weights

Model Configuration	Weights (w_C, w_D, w_O)	Avg. Utility	Accuracy	Precision	Recall	F1-Score
Fixed utility weights (Random Forest)	(0.33, 0.33, 0.34)	0.1258	89.42%	0.897	0.894	0.893
Fixed utility weights (Random Forest)	(0.80, 0.10, 0.10)	0.3178	89.37%	0.896	0.893	0.893
Fixed utility weights (Random Forest)	(0.15, 0.70, 0.15)	0.0512	88.75%	0.888	0.887	0.887
Optimized utility weights (Random Forest)	(0.7, 0.2, 0.1)	0.3548	89.71%	0.899	0.897	0.896
Optimized utility weights (XgBoost)	(0.7, 0.2, 0.1)	0.3548	92.94%	0.930	0.929	0.929
Optimized utility weights (Decision Tree)	(0.7, 0.2, 0.1)	0.3548	88.84%	0.889	0.888	0.888

weight baselines. The optimized utility weights (RF) achieves the highest accuracy (89.71%), average utility (0.3548), F1-score (0.896), precision (0.899), and recall (0.897), demonstrating that systematically discovered weights yield consistently superior performance over all fixed weights (RF). Further, Table I also presents the three-way comparison of Random Forest, Decision Tree, and XGBoost under the optimized label set. XGBoost achieves the highest accuracy (92.94%), slightly above Random Forest (89.71%) and Decision Tree (88.84%). Despite XGBoost's marginally higher accuracy, the Random Forest offers superior interpretability through Gini-based feature importance and lower deployment complexity, making it preferable for on-vehicle edge inference.

VI. CONCLUSION

This paper proposed a machine learning-based adaptive link-selection framework for VLC–RF vehicular communication under realistic weather impairments, extending conventional designs to a three-class decision space comprising RF, VLC, and Link-Aggregated modes. A multi-objective utility function balancing throughput, delay, and outage probability was developed, with optimal weights (w_C, w_D, w_O) = (0.7, 0.2, 0.1) obtained via 10-fold cross-validation. The optimized Random Forest classifier achieved 89.71% accuracy, and comparative results with Decision Tree and XGBoost validated the effectiveness of the proposed approach for intelligent link selection. Under diverse weather conditions, the Link-Aggregated mode consistently outperformed standalone RF and VLC by achieving higher utility, lower delay, and improved reliability, particularly in dense fog where it provided up to 38.6% gain in utility and nearly one order of magnitude delay reduction at longer distances. Feature analysis further confirmed fog attenuation, inter-vehicle distance, and weather class as the dominant decision factors, demonstrating the robustness of the proposed framework for weather-resilient, low-latency vehicular platooning. Future work will investigate SHAP-guided explainable machine learning for feature attribution and adaptive utility optimization to further enhance decision interpretability, robustness, and real-time link selection in highly dynamic vehicular environments.

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